Dissertation Outline

LLM-Enhanced Dynamic Graph Networks for Conversion Rate Prediction: Integrating Large Language Model Representations to Mitigate Delayed Feedback Bias in Recommender Systems

UCL · MSc Knowledge, Information and Data Science

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Total estimated length: 15,000–16,000 words

Text / Sequences → LLM Encoder → Embeddings → Dynamic Graph → GNN + Time-Decay → CVR Score

Ch 1 Introduction

~1,500 words

■ Background & motivation

CVR prediction in online advertising; the gap between CTR and CVR; industrial scale and business impact.

■ Problem statement

Delayed feedback causes false-negative bias; existing GNN methods ignore semantic richness of item/user features.

■ Research questions [see RQs below]

■ Contributions

Novel LLM+GNN hybrid framework; ablation study isolating LLM contribution; public dataset benchmarking.

■ Thesis structure

Research Questions

RQ1 Can LLM-generated semantic embeddings improve CVR prediction over ID-based features in dynamic graph networks?

RQ2 Does integrating LLM representations reduce the adverse effects of delayed feedback bias?

RQ3 What is the computational trade-off between LLM-enhanced and traditional GNN approaches?

Ch 2 Literature Review

~3,000 words

■ CVR prediction fundamentals

DeepFM, DIN, DIEN — evolution from shallow to deep models; feature interaction paradigms.

■ Delayed feedback problem [CORE]

ES-DFM, DEFER, FSIW — existing approaches and their limitations.

■ GNNs in recommender systems

Dynamic information networks; temporal graph modelling; time-decay mechanisms.

■ LLMs for recommendation [CORE]

UniSRec, RLMRec, TALLRec — LLM as encoder vs. LLM as ranker; semantic feature learning.

■ Research gap

No prior work combines LLM semantic encoding with dynamic GNN specifically for delayed feedback CVR.

Coverage: ~15–20 key papers across the four sub-themes above.

Ch 3 Methodology

~3,500 words

■ Overall framework

Three-stage pipeline: LLM encoding → dynamic graph construction → GNN-based CVR prediction. Include architecture diagram.

■ Stage 1 · LLM semantic encoding [CORE]

Pre-trained model selection (BERT / sentence-transformers); encoding item text & user behaviour sequences; embedding dimensionality and normalisation.

■ Stage 2 · Dynamic graph construction

Node definition (users, items); edge definition (interactions with timestamps); time-sliced graph snapshots; incorporating LLM embeddings as node features.

■ Stage 3 · GNN-based prediction [CORE]

Graph convolution layers; time-decay weighting function for delayed feedback; aggregation and output MLP for CVR score.

■ Training strategy

Loss function (binary cross-entropy); handling label delay in mini-batches; optimiser (Adam); regularisation.

■ Theoretical justification

Why semantic features from LLM complement structural features from GNN; information-theoretic argument.

Pipeline summary: Text/Sequences → LLM Encoder → Embeddings → Dynamic Graph Nodes → GNN + Time-Decay → CVR Score

Ch 4 Experimental Setup

~2,000 words

■ Datasets

Amazon Review (primary — rich text features); AliCCP (secondary — real-world CVR with delayed labels); preprocessing steps.

■ Baselines

Traditional: DeepFM, DIN; GNN-based: DGNN / SURGE; LLM-only: sentence-BERT features + MLP (no graph structure).

■ Evaluation metrics

AUC (primary); Log Loss; GAUC (group-level AUC); inference latency (ms/sample).

■ Implementation details

Python 3.10, PyTorch 2.x, PyTorch Geometric, HuggingFace Transformers; GPU: Colab A100; hyperparameter search space.

■ Reproducibility

Random seed fixation; dataset splits (7:1:2); public code release plan.

Ch 5 Results & Analysis

~3,000 words

■ Main results

Comparison table: proposed model vs. all baselines on AUC and Log Loss across both datasets.

■ Ablation study [CORE]

Variant 1: remove LLM embeddings (ID features only); Variant 2: remove time-decay (standard GNN); Variant 3: remove graph (LLM + MLP only) — isolates each component.

■ LLM encoder comparison

BERT-base vs. sentence-transformers vs. domain-tuned variant; effect on downstream AUC.

■ Delayed feedback sensitivity

Performance under varying feedback delay windows (1h, 6h, 24h, 7d); does LLM help more under longer delays?

■ Efficiency analysis

Training time, inference latency, memory footprint vs. baselines; Pareto trade-off chart.

■ Case study / qualitative analysis

Example items where LLM embedding changes the prediction; attention visualisation on graph.

Ch 6 Discussion

~1,500 words

- **Interpretation of findings**

  Answers to each RQ; when does LLM help and when doesn't it?

- **Comparison with related work**

  How results relate to UniSRec, RLMRec findings; where this work advances the field.

- **Limitations**

  Computational cost of LLM inference; dependence on text quality in dataset; single-domain evaluation.

- **Future work**

  Fine-tuning LLM on domain data; cross-domain transfer; real-time serving optimisation.

Ch 7 Conclusion

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- **Summary of contributions**
- **Broader implications for LLM-enhanced recommendation**
- **Final remarks**

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